Path 2 — Only MRI + Masks

Two solid options are available when elastography data (MRE/SWE) are not present:

2A) Mass-effect Inverse Fitting (Patient-Specific)

Idea: Derive a target displacement field from image registration, then fit material parameters (and/or growth pressure) to minimize the difference with FE-predicted displacement.

Workflow:.

1. **Create a target displacement field:**
   - Deformably register a healthy/atlas brain to the patient scan, or register pre-operative to follow-up scans.
   - Extract the displacement field $u_{\mathrm{MRI}}x$ in the brain volume (excluding the skull).
2. **Finite Element (FE) Setup:**
   - Mesh: include tumor and surrounding brain (GM, WM, CSF).
   - Boundary conditions: constrain nodes at the brain–skull interface (or use spring/sliding contact). Treat ventricles/CSF as compliant with low stiffness.
   - Optional: include growth pressure or volumetric expansion in the enhancing tumor (ET) to mimic tumor driving forces.
3. **Unknowns:** $\theta = \{E_{\mathrm{ET}}, E_{\mathrm{NCR}}, E_{\mathrm{ED}}, E_{\mathrm{WM}}, E_{\mathrm{GM}}\}$ (regional elastic moduli) and, if modeled, growth pressure p_g.
4. **Objective Function:**

$$J\theta = \|u_{\mathrm{FE}}\theta - u_{\mathrm{MRI}}\|^2 \;\; \lambda\|\theta\|^2$$

5. **Optimization:**
   - Use CMA-ES or another gradient-free optimizer.
   - Bounds: $E \in 1, 20$ kPa, $p_g \in 0, 5$ kPa.
6. **Reporting:** Provide fitted E/G values per region and confidence intervals (via bootstrap over landmarks/ROIs).

For similar inversion strategies in neuro-FE, see model-based elastography and inverse FE surveys.

2B) Literature-Prior Forward Analysis (Fast & Transparent)

If only plausible values are needed for simulations or exploratory studies:

Workflow:.

1. **Assign prior values (kPa shear modulus G):**
   - WM: $G \approx 1.5$–2.5
   - GM: $G \approx 1.0$–2.0
   - ET: $G \approx 1$–3
   - NCR/NET: softer (< 1)
   - ED: $G \approx 0.5$–1.5
2. **Set Poisson's ratio:** $\nu = 0.45$–0.49 (use mixed elements to avoid locking).
3. **Convert to E:**

$$E = 2G1 \;\; \nu \quad \nu = 0.49 \Rightarrow E \approx 2.98G$$

4. **Run virtual tests:** Apply 1% uniaxial compression or shear. Report effective E, G, and stress/strain maps per region.
5. **Validation:** Check mesh-independence and sensitivity to boundary conditions. Adjust priors within literature ranges.

Step 4 — Boundary Conditions & Loads (All Paths)

- Brain–skull: fix outer surface displacement or use spring foundations to mimic meninges/sliding.
- CSF/ventricles: model as very compliant; if poroelastic, couple to fluid pressure.
- Loads:
  - Calibration: apply small displacements (1%) on one face, fix the opposite.
  - Physiologic: add growth pressure in ET or traction to mimic surgical retraction.

Step 5 — Outputs to Compute

For each region (ET, NCR/NET, ED, WM, GM):

- Primary: G (kPa), E (kPa), ν.
- Hyperelastic: μ and K (or Ogden/Mooney–Rivlin parameters).
- Viscoelastic: Prony terms g_i, τ_i; report storage $G'\omega$ and loss $G''\omega$.
- Poroelastic: permeability k (mm^2/Pa·s), porosity ϕ, interstitial pressure p.
- Maps: stress, strain, and effective stiffness visualizations.

Quick Cookbook (Example)

From MRE speed map to FE material:.

1. Resample $c_s x$ onto mesh nodes/cells.
2. Compute shear modulus: $\mu x = \rho c_s x^2$ (with $\rho = 1040$ kg/m^3).
3. Set $\nu = 0.49$, then $Ex = 2\mu 1 \;\; \nu \approx 3\mu$.
4. Average by region; export a materials table:

```
Region, E_kPa, nu
ET,   6.2, 0.49
NCR,  2.1, 0.49
ED,   3.0, 0.49
WM,   7.0, 0.49
GM,   4.5, 0.49
```

Inverse Mass-effect (Pseudo-code):.

```
# \theta = [E_ET, E_NCR, E_ED, E_WM, E_GM, p_g]  # kPa
bounds = [(1,20)]*5 + [(0,5)]
theta = init_from_priors()
for iter in range(max_iter):
    write_FE_input(mesh, regions, theta, nu=0.49, growth_pressure='ET')
    run_solver()
    u_FE = read_displacement_field()
    loss = ||u_FE - u_MRI||^2 + \lemda*||theta||^2
    theta = optimizer.step(theta, loss, bounds)
report(theta)
```

Recommended Tools

- Meshing: Gmsh, TetGen, iso2mesh (Matlab/Python).
- Solvers (open-source): FEBio, FEniCSx/Dolfinx, SOFA, Elmer.
- Registration: ANTs, SimpleITK, MONAI.
- MRE processing: vendor pipelines or open-source implementations; follow community guidelines.

Validation & Reporting (Must-do)

- **Mesh convergence:** refine elements (especially in tumor/edema) until moduli change $< 5\%$.
- **Boundary sensitivity:** test fixed vs. spring skull constraints.
- **Uncertainty:** bootstrap over ROIs or perturb u_{MRI} with noise for CIs.
- **Context:** report in relation to literature (gliomas generally softer than healthy brain, with heterogeneous stiff foci possible).

Summary (TL;DR)

- **Have MRE/SWE?** Use Path 1 (direct $c_s \to \mu \to E$).
- **Only MRI+Masks?**
  - Patient-specific: Path 2A (inverse mass-effect).
  - Pragmatic/fast: Path 2B (literature priors + sensitivity analysis).